

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's per-species, per-place reliability —
for detection and association — from label-free distances

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Declaration of Originality.

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Abstract

Promptable trackers such as SAM3 segment and track any named animal in a video with no training on that species, yet no rule tells a practitioner *in advance* whether the model is reliable for a given animal in a given place. This dissertation builds and validates a *predictive* rule for that reliability on the SA-FARI benchmark, whose splits are disjoint by species and location. We measure SAM3’s zero-shot promptable-tracking accuracy with the official evaluator, decompose it into detection (**pDetA**) and association (**pAssA**), and test whether each can be predicted from four label-free distances (taxonomic, visual, environment, temporal). *[TODO: headline result + validated out-of-sample performance.]*

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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM 3 [3] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM 3’s zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)

the label-free distances carry no predictive information about pDetA/pAssA — transfer is not distance-driven.

H1 (detection)

pDetA falls with species novelty (taxonomic + visual distance).

H2 (association)

pAssA falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 would itself be an informative result, motivating a pivot from distance-based prediction to representational probing (does SAM 3 encode taxonomy?).

1.4 Contributions

The contribution is, to our knowledge, the first predictable-transfer study for a promptable video tracker, decomposed into two predictors (detection vs association), on SA-FARI’s species $\times$ place $\times$ time natural experiment — yielding a deployment reliability estimator.

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

2.1 Promptable segmentation and tracking

We introduce promptable trackers and SAM3 [3], the model whose transfer we study.

2.2 The tracking metric and its decomposition

The HOTA family [4] decomposes tracking accuracy into detection and association components — the split that is the intellectual core of this work.

2.3 Predictable out-of-distribution generalisation

“Accuracy on the Line” [5] and “Agreement on the Line” [1] show OOD accuracy is predictable (and label-free) for *classifiers*; no analogue exists for promptable video trackers.

2.4 What should predict transfer

Vision foundation models carry taxonomic structure [8] (motivating taxonomic/visual distance), and camera-trap recognition degrades at new locations [2] (motivating environment distance). We embed animals/scenes with DINOv2 [6], with CLIP [7] as a robustness variant.

2.5 The SA-FARI benchmark

SA-FARI [9] spans 99 species (7-level taxonomy), 741 locations across four continents, with train/test disjoint by species *and* location — a controlled transfer probe.

2.6 Summary and gap

Chapter 3

Data and Methodology

3.1 Dataset and the analysis unit

We aggregate scores to a *cell* — a (species, location, time) group — carrying support counts.

3.2 Measuring transfer

SAM3 [3] is run frozen over the test split and scored with the official evaluator (the metric is never re-implemented), giving per-cell pDetA/pAssA/pHOTA [4].

3.3 The frozen reference (leakage firewall)

The seen set is frozen once; every distance is computed against the training split only, so no test information can leak into a feature.

3.4 Label-free distance features

3.4.1 Taxonomic distance

3.4.2 Visual distance

3.4.3 Environment distance

3.4.4 Temporal gap

3.4.5 SAM 3 familiarity proxy

3.5 Modelling the law

Each bounded target is modelled with a support-weighted beta regression (with a $\log(n_{\text{frames}})$ covariate), variance is attributed by dominance analysis, and the predictor is validated out-of-sample by holding out whole species and whole locations.

3.6 Reproducibility and constraints

Chapter 4

Results and Evaluation

4.1 Measurement

4.2 The fitted law

4.3 Variance partitioning and the decomposition test

4.4 Out-of-sample validation

4.5 Ablations and robustness

Chapter 5

Discussion

5.1 Interpretation

5.2 A deployment reliability estimator

5.3 Limitations and threats to validity

5.4 Ethical considerations

Chapter 6

Conclusion

6.1 Summary of findings

6.2 Contributions

6.3 Future work

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Appendix A

Reproducibility

Appendix B

Supplementary Results